

I Thought We Had 2,000 Tests. We Had 2 Monoliths and a Race Condition.

The honest story of refactoring 4,000 lines of AI-generated tests and the bug we found by accident.

LINES DELETED

~4,000

2 monoliths → 23 modules

TESTS RUNNING

1,200+

292 unique × 4 browsers

BUGS FOUND

1

Race — 1 line fix

AI MODELS

17

5 still free · 7 gone

Two Files. Nearly 4,000 Lines. No Structure.

AI generated fast. AI generated without architecture.

Before

3,908 lines

- ✗ UI monolith — 2,349 lines
- ✗ API monolith — 1,559 lines
- ✗ 15 duplicated API blocks
- ✗ No shared fixtures



After

23 modules

- ✓ 9 API modules (auth, posts...)
- ✓ 14 UI modules (profile, search...)
- ✓ Shared fixtures.ts
- ✓ CI smoke in 3 minutes

HOW WE SPLIT IT

Refactoring AI Code With More AI Was Surprisingly Hard

Every tool hit a wall. The human had to finish it.

Tool	Result	Limit
Cursor (Hobby)	✅ 1 file done	50 requests — spent
MiniMax M2.5 (OpenCode)	✅ ~20 files refactored	Promo ended
Cline (OpenRouter)	❌ Failed	Couldn't handle >2,000 lines
Victor (human)	✅ Reviewed & approved	Final say

FOUND BY ACCIDENT

A Race Condition Lived in Production for 2 Months. Tests Never Caught It.

We found it because refactoring forced us to *read* the code.

ROOT CAUSE

`datetime.now()`

Same microsecond → same JWT

`unique_violation` in PostgreSQL

THE FIX

`jti: uuid.uuid4()`

One line in the token payload

Every token now unique

BUG AGE

2 months

Lived since day 1

TRIGGER

4× browsers

Parallel runs exposed it

DETECTION

Refactoring

Reading code, not tests

We Didn't Write a Single New Test in May. It Was Our Most Productive Month.

Adding tests is easy. Deleting ~4,000 lines is where the value was.

Phase 1 (April)

- 750+ tests built
- API coverage: 94%
- PBT coverage: 100%
- 2 monoliths ~4,000 lines
- No CI pipeline

Phase 2 (May)

- ✓ Zero new tests written
- ✓ ~4,000 lines deleted
- ✓ 2 → 23 modular files
- ✓ CI smoke: 12 endpoints / 3 min
- ✓ 1 bug found (race condition)

"Sometimes the best test improvement is deleting 4,000 lines and starting cleaner."

17 Models in 2 Months. 5 Still Free. 7 Already Gone.

The treadmill is real. Don't rely on one provider.

● Active — 6 tools

OpenCode × 3 (glm-5, minimax-m2.7, gpt-5-nano)
GitHub Copilot · Ollama qwen2.5:3b · Groq Llama 3.3 70B

ALL FREE

● Gone — 7 tools

MiniMax M2.5 (promo) · Cursor Hobby (50 req)
Cline · Gemini API (429) · qwen2.5:14b (timeout)
Continue.dev · Big Pickle · Mimo V2 Pro/Omni
FREE TIER ENDED



Free AI is a treadmill

Keep moving or get left behind. Maintain a stack of 4–5 free tools across providers and be ready to swap any week.

Three Honest Takeaways

1

AI amplifies your prompt quality

"Write tests" generated 4,000 lines of tech debt. A specific prompt with modular structure would have saved Phase 2 entirely.

2

Refactoring finds bugs that testing misses

A race condition lived for 2 months. 1,000+ tests ran daily. None caught it. Reading the code found it in 5 minutes.

3

Free AI is a treadmill, not a foundation

17 tools in 2 months. 7 are gone. 5 remain. Diversify, rotate, never depend on a single provider.

"The biggest lesson? Sometimes the best test improvement is deleting 4,000 lines."

This Is Only Half the Stack.

The monoliths are gone. Architecture is documented.
Now we expand across 5 languages.



Go Backend

11 tests, 2 packages — users & follows.
Race detection, table-driven tests, error handling.



C# + Fuzzer

41 tests, 5 modules — Fuzzer, Properties,
Schema, Race, Metamorphic. PBT on .NET.



Mutation Testing

API + DB mutations + Chaos testing.
Bugs found: BUG-005 (XSS), BUG-006 (negative likes).

5 languages · TypeScript · Go · C# · Python · k6
7 bugs found to date · 5 architecture docs · 3 diagrams

YOUR TURN

What's the messiest test suite AI has helped you create?

Share your story in the comments. How did you clean it up —
or are you still living with the debt?

~4,000

Lines deleted

\$0

Budget

1 bug

Found by accident



Victor P.

QA Automation Engineer · Building in public
victor-2026.github.io/qa-automation-playwright